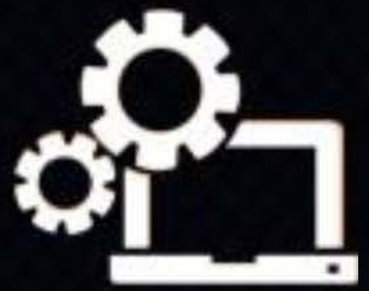
CS & IT ENGINEERING

DIGITAL LOGIC

Combinational Circuit





By - CHANDAN SIR

Lecture No. 03

Recap of Previous Lecture



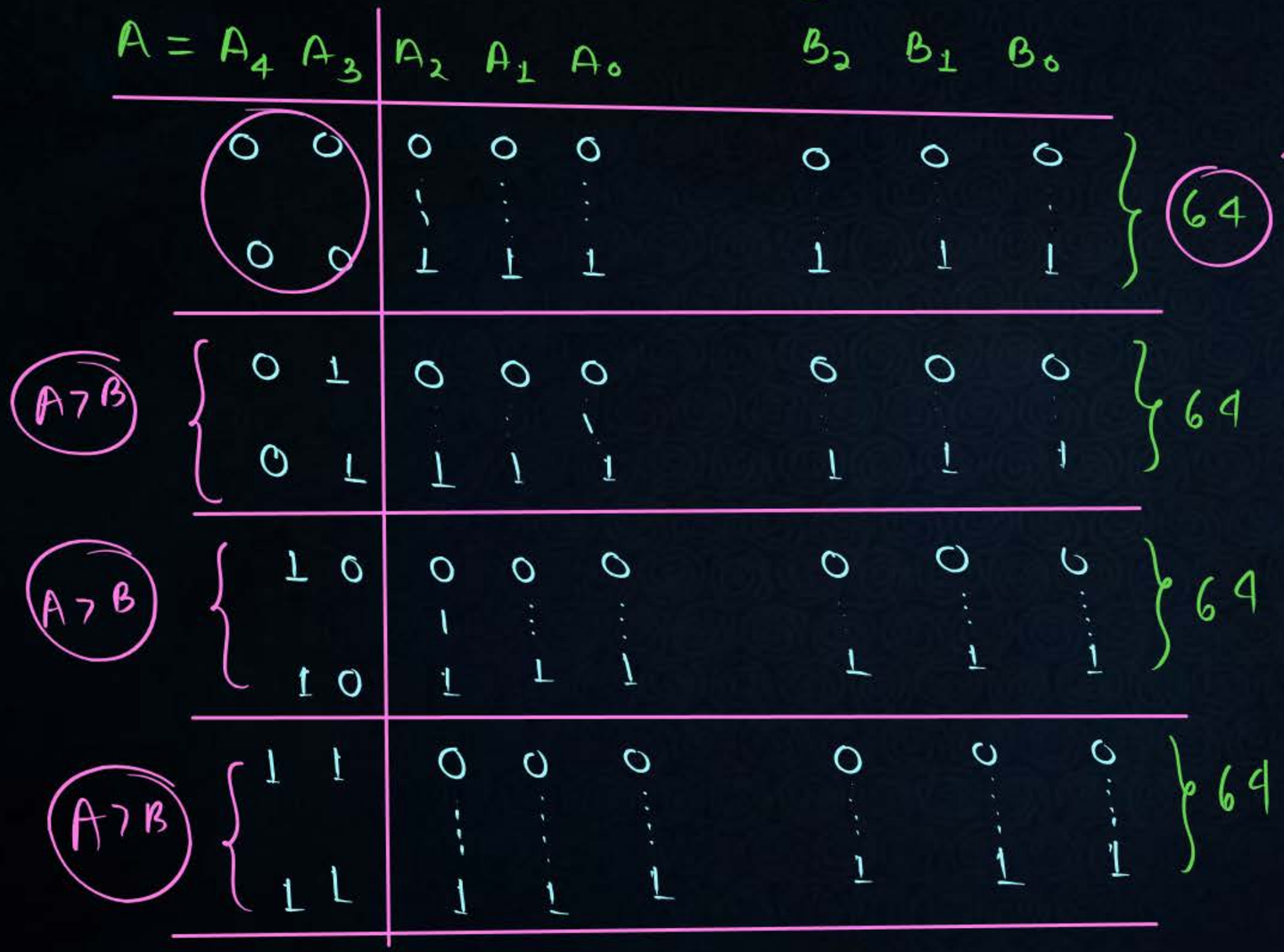
Comparator

Topics to be Covered



MULTIPLEXER

- MULTIPLEXER
- QUESTIONS PRACTICE
- DISCUSSION



3 bit comparator 64 /

A=B ⇒ 8

A>B ⇒ 28

A<B ⇒ 28

A>B ⇒ 28+64+64+64 = 220

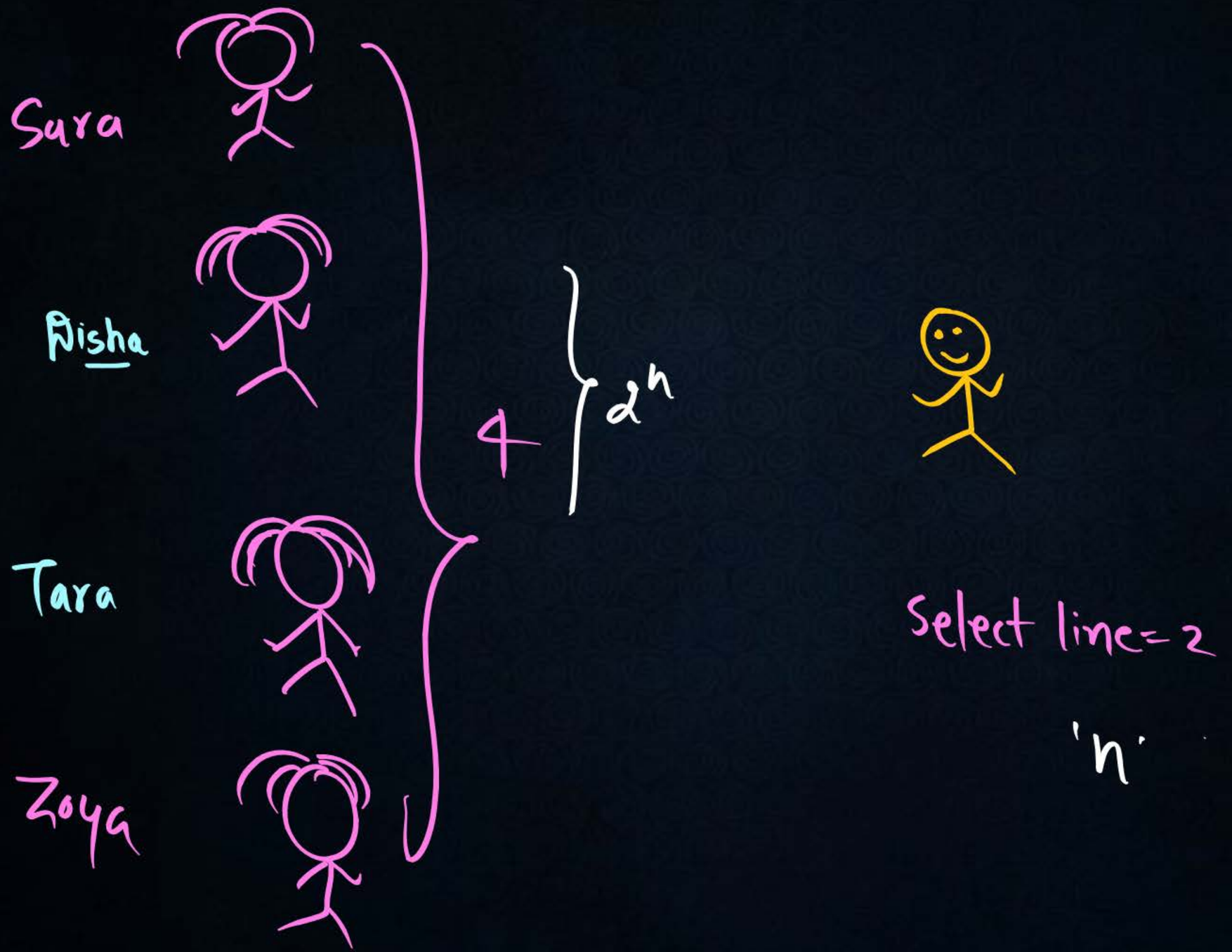
A<B ⇒ 28

A=B ⇒ 8

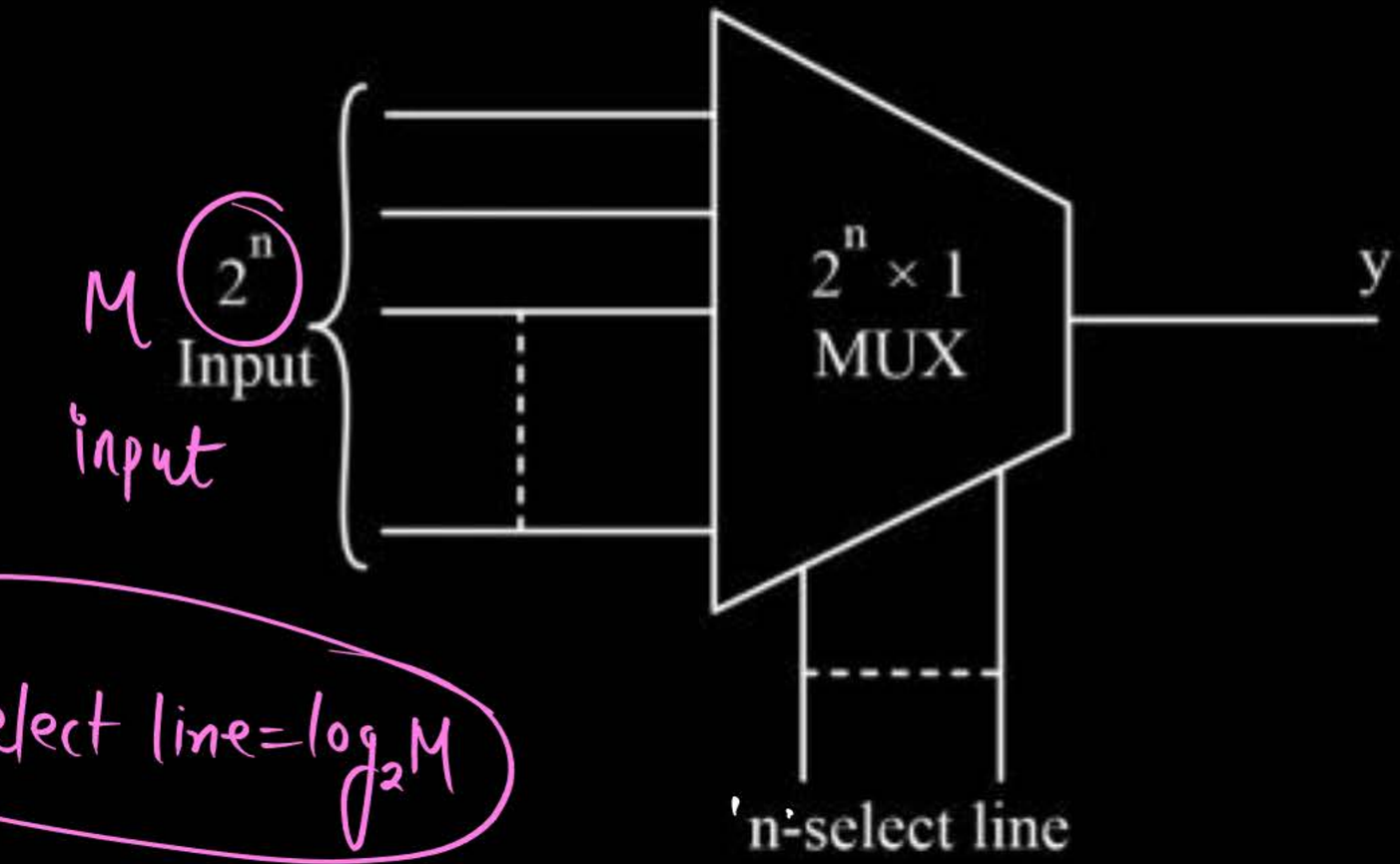


Topic : MULTIPLEXER

- many to one converter
- universal Logic
- AND - OR Logic


Sara
Nisha
Tara
Zoya
4
2^n
Select line = 2
'n'

Topic : MULTIPLEXER



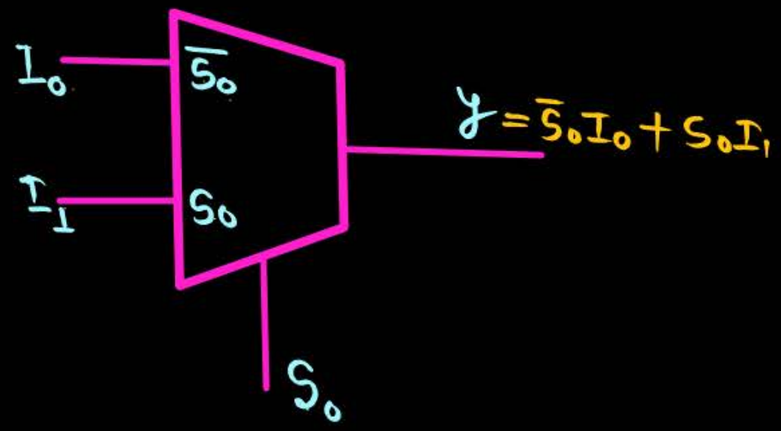
no. of select line = $\log_2 M$



Topic : Designing MUX

Design a 2 × 1 MUX

Step 1 :→



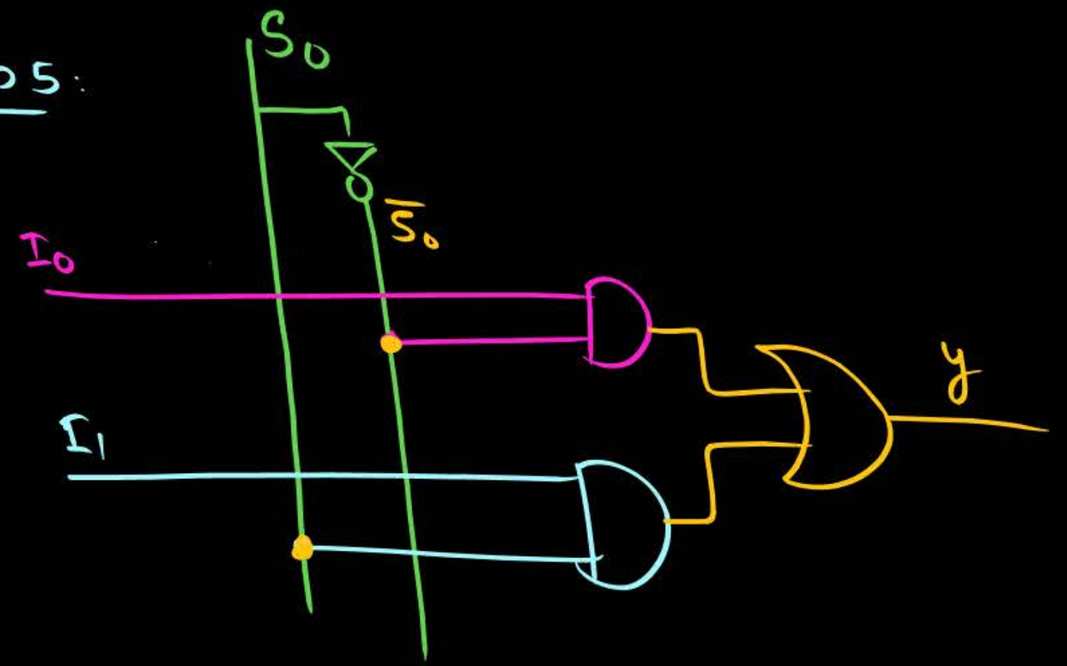
Step 2:

S_0	y
0	I_0
1	I_1

Step 3: $y = \underline{\bar{S}_0 I_0} + \underline{S_0 I_1}$

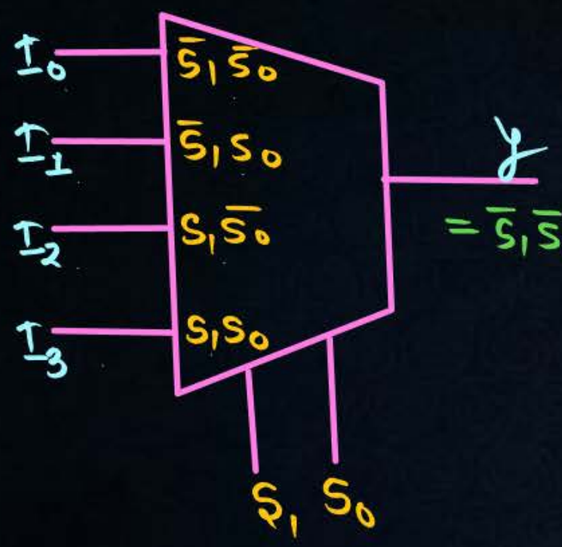
Step 4 :→

Step 5:



Design a 4x1 MUX ?

Step 1:



$$= \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$$

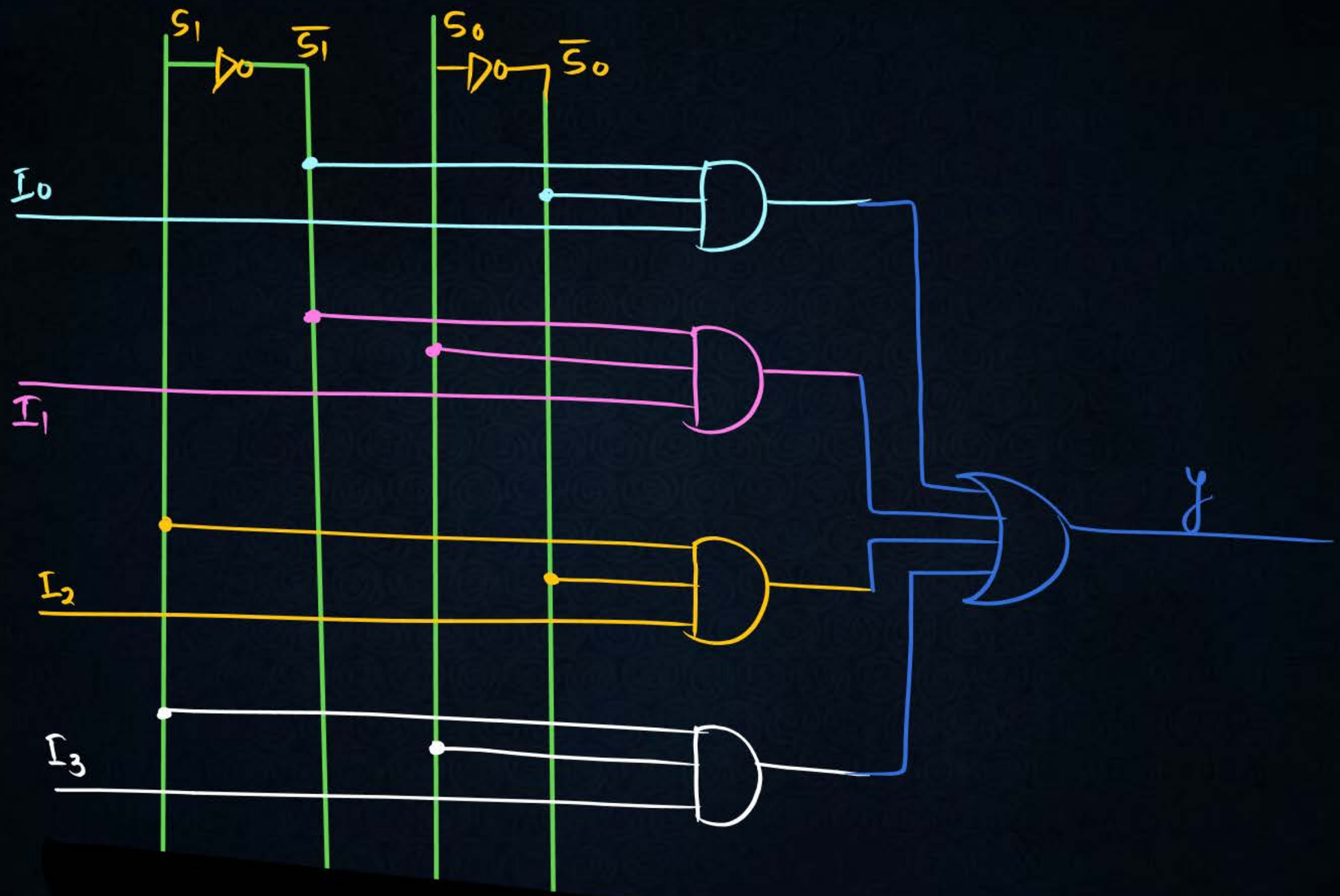
Step 2:

S_1	S_0	y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Step 3: $\rightarrow$ $y = \bar{S}_1\bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1\bar{S}_0 I_2 + S_1 S_0 I_3$

Step 4: already minimized

Step 5 :→



HW

Design a 8x1 MUX ?

Design a 16x1 MUX ?



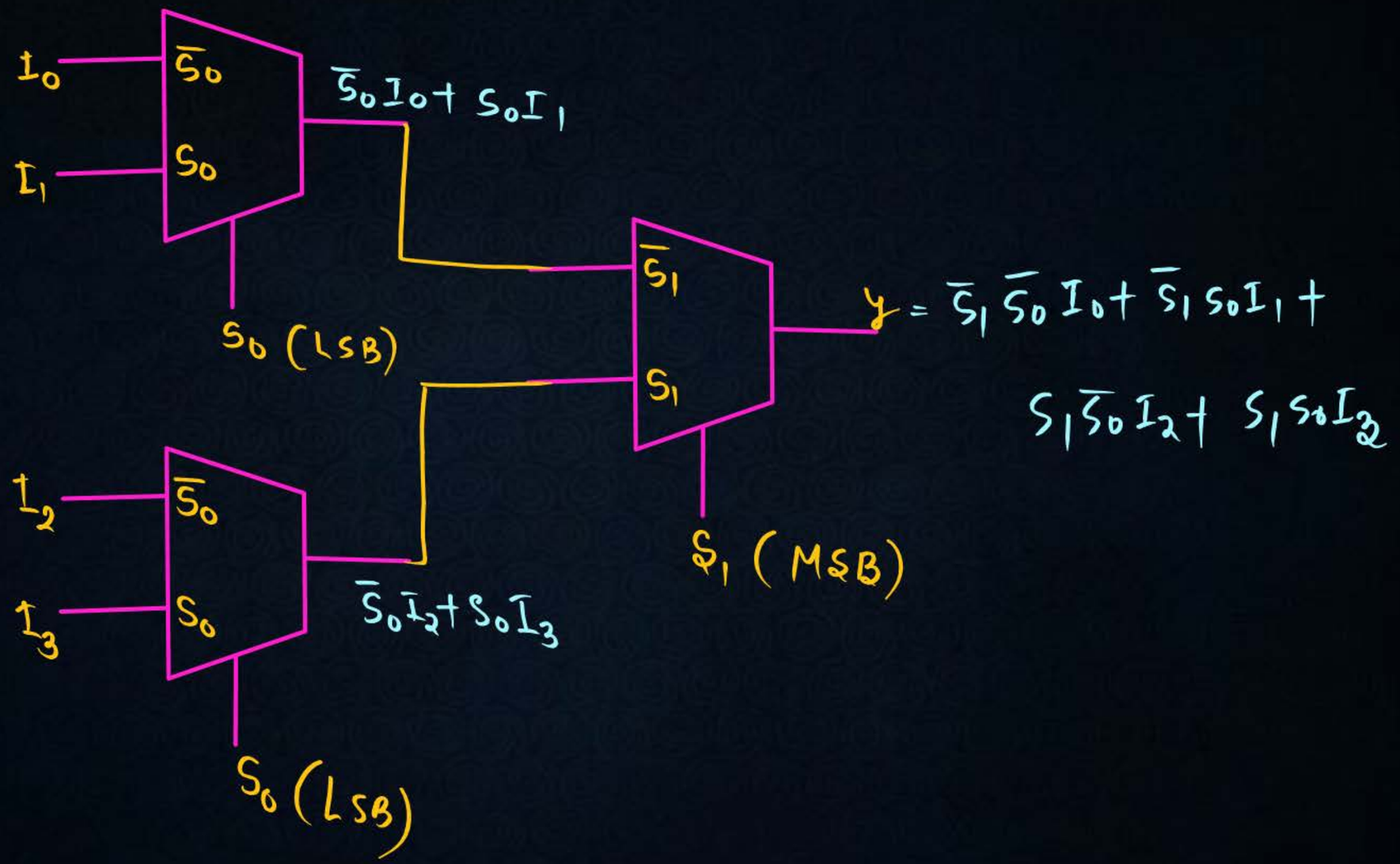
Type-1 : Designing a higher order Mux by Using Lower Order Mux

#Q. Design 4 × 1 MUX using 2 × 1 MUX.

$\frac{4}{2} + \frac{2}{2}$

2 × 1 MUX → 4 × 1 MUX

$2 + 1 = 3$


I_0
I_1
$\bar{S}_0$
S_0
$\bar{S}_0 I_0 + S_0 I_1$
S_0 (LSB)
I_2
I_3
$\bar{S}_0$
S_0
$\bar{S}_0 I_2 + S_0 I_3$
S_0 (LSB)
$\bar{S}_1$
S_1
S_1 (MSB)
$y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$

Q 2×1 MUX $\xrightarrow{\frac{8}{2}+\frac{4}{2}+\frac{2}{2}}$ 8×1 MUX

$4+2+1=7$

Ans

Q 2×1 MUX $\xrightarrow{\frac{16}{2}+\frac{8}{2}+\frac{4}{2}+\frac{2}{2}}$ 16×1 MUX

$8+4+2+1=15$

2×1 MUX $\xrightarrow{2^n-1}$ $2^n\times1$ MUX

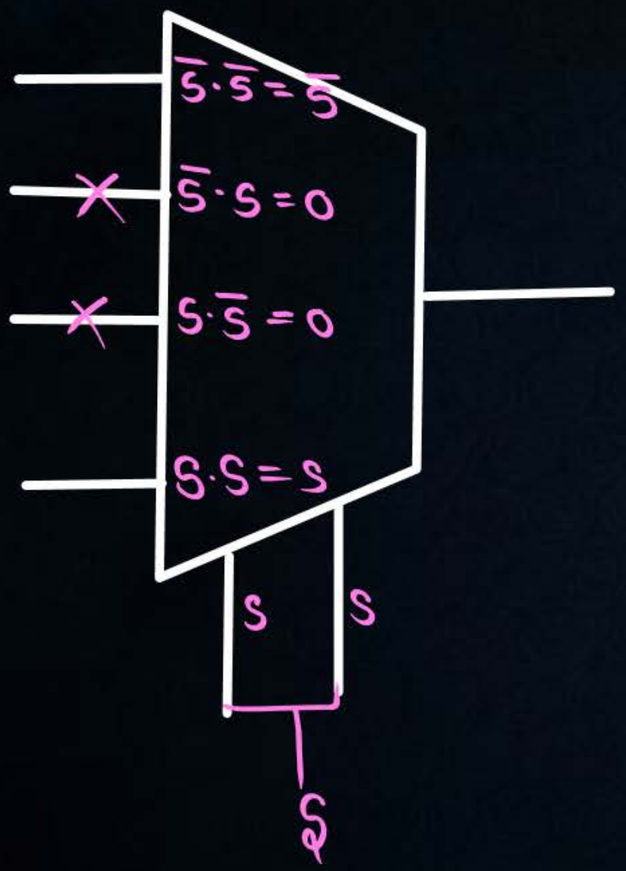
Q 4x1 MUX $\xrightarrow{\frac{16}{4}+\frac{4}{4}}$ 16x1 MUX

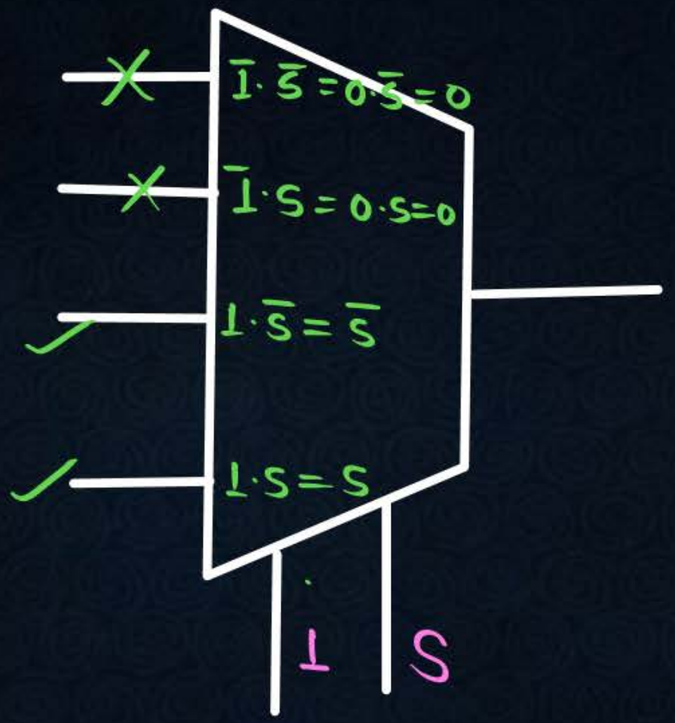
$4+1=5$

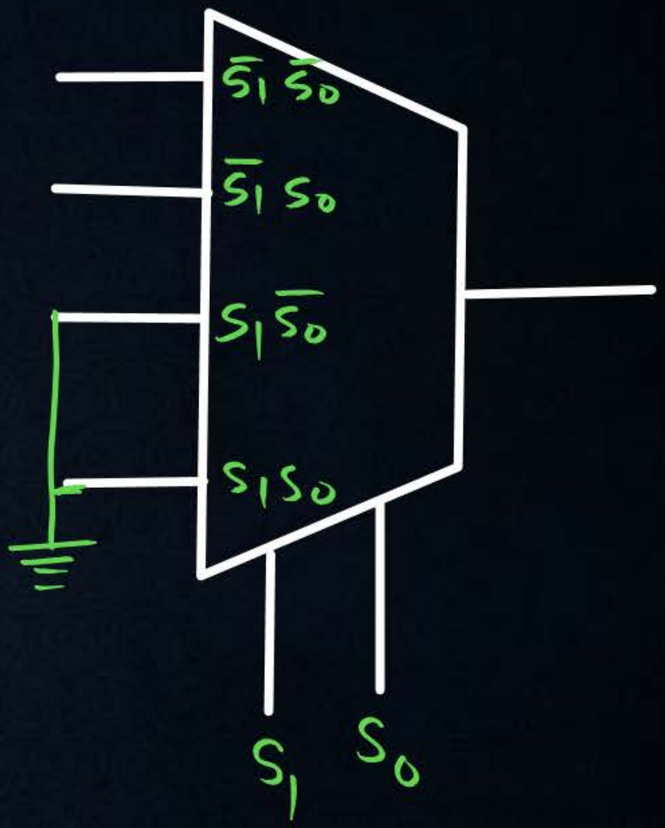
Q 4x1 MUX $\xrightarrow{\frac{64}{4}+\frac{16}{4}+\frac{4}{4}}$ 64x1 MUX

$16+4+1=21$

Ans

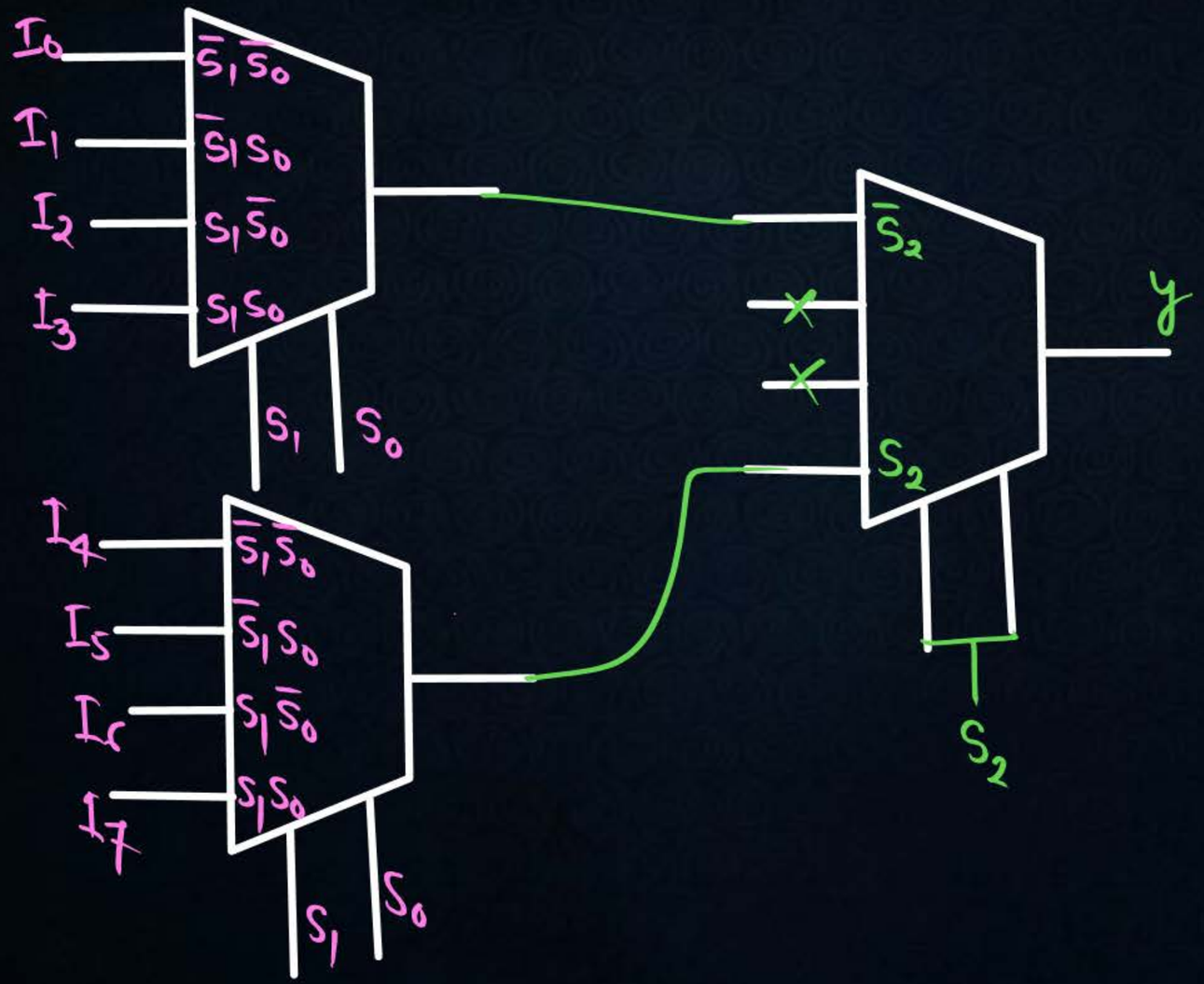

$\bar{S}\cdot\bar{S}=\bar{S}$
$\bar{S}\cdot S=0$
$S\cdot\bar{S}=0$
$S\cdot S=S$
S
S
S


$\bar{1}\cdot\bar{S}=0\cdot\bar{S}=0$
$\bar{1}\cdot S=0\cdot S=0$
$1\cdot\bar{S}=\bar{S}$
$1\cdot S=S$
1
S


$\bar{S}_1\bar{S}_0$
$\bar{S}_1 S_0$
$S_1\bar{S}_0$
$S_1 S_0$
S_1
S_0

Q 4×1 MUX $\xrightarrow{\frac{8}{4}+\frac{2^{2}}{4}}$ 8×1 MUX

$2+1=3$



Q ≡ 8 x 1 MUX $\xrightarrow{\frac{64}{8} + \frac{8}{8}}$ 64 x 1 MUX

$8 + 1 = 9$

Q. 8 x 1 MUX $\xrightarrow{\frac{32}{8} + \frac{4}{8}}$ 32 x 1 MUX

$4 + 1 = 5$

Q 4x1 MUX $\xrightarrow{\frac{32}{4}+\frac{8}{4}+\frac{2}{4}}$ 32x1 MUX

$8+2+1=11$

Q 16x1 MUX $\xrightarrow{\frac{256}{16}+\frac{16}{16}}$ 256x1 MUX

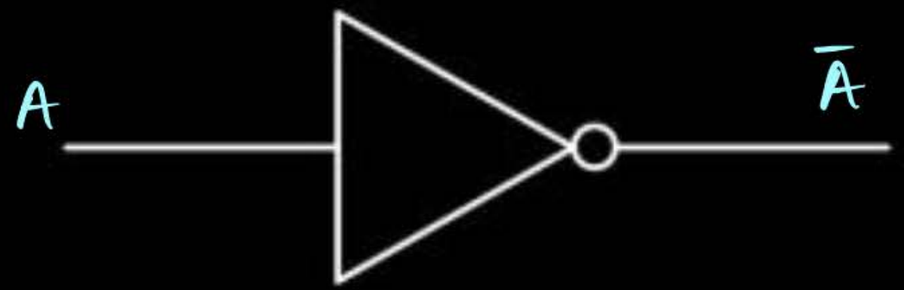
$16+1=17$

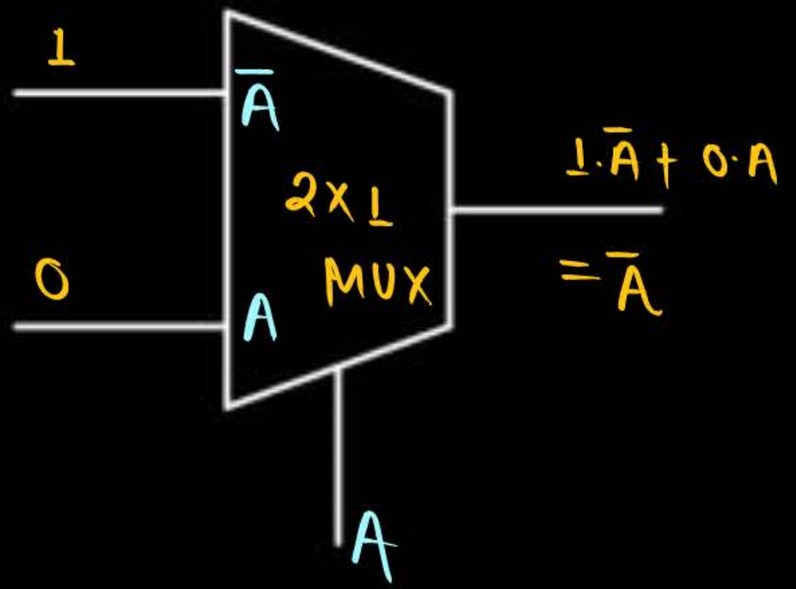
Ans



Type-2 : MUX as a Universal Logic

1. Not GATE

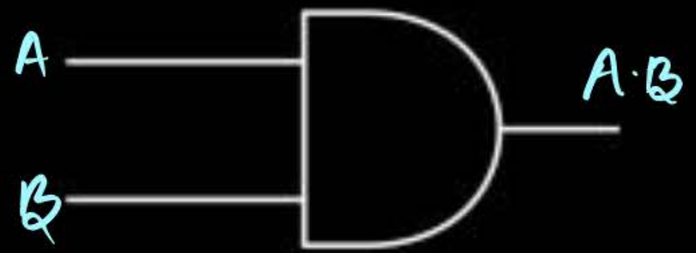


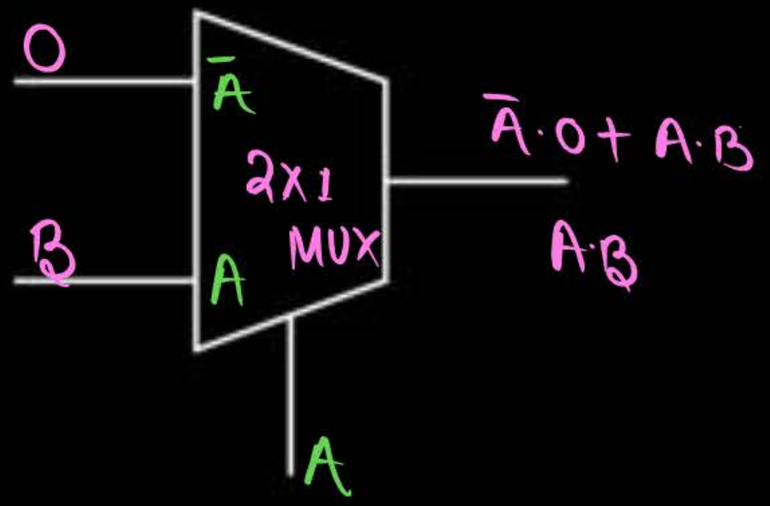




Type-2 : MUX as a Universal Logic

2. AND GATE

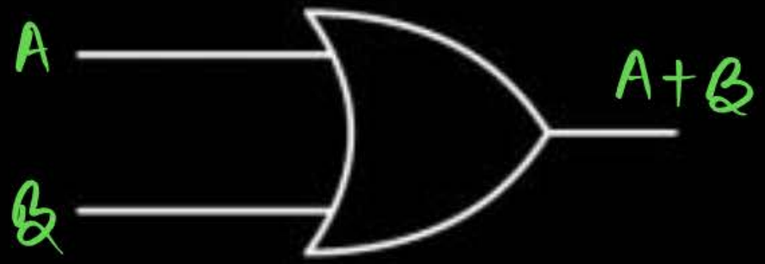


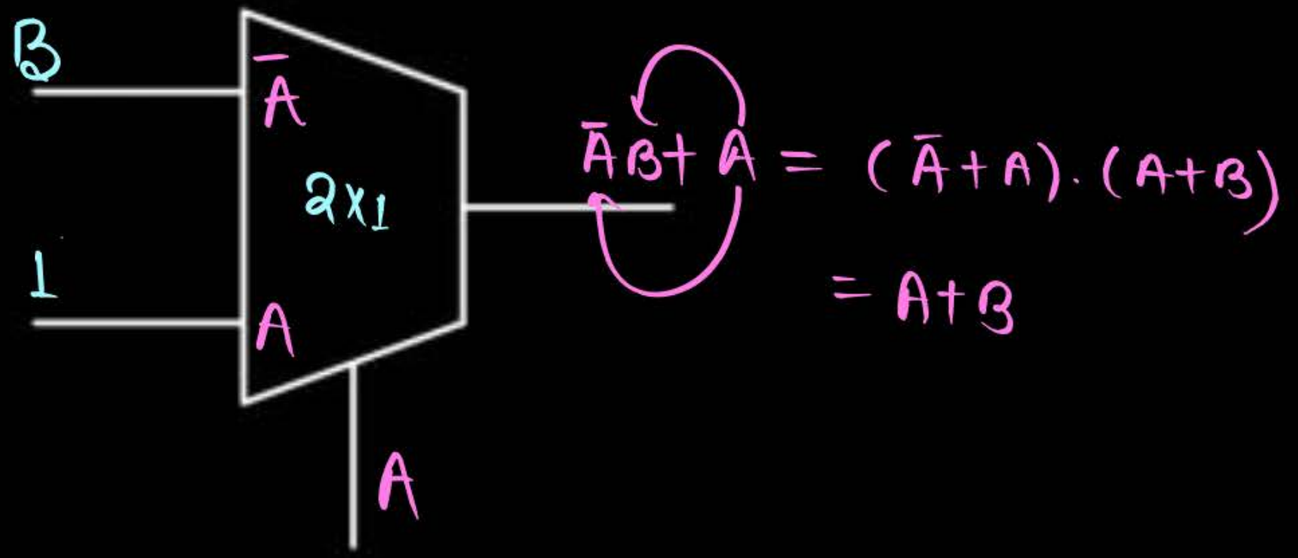




Type-2 : MUX as a Universal Logic

3. OR GATE

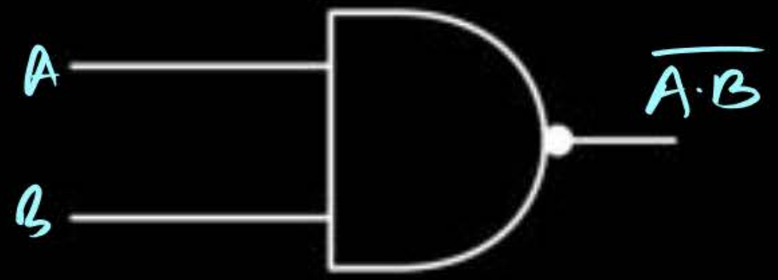


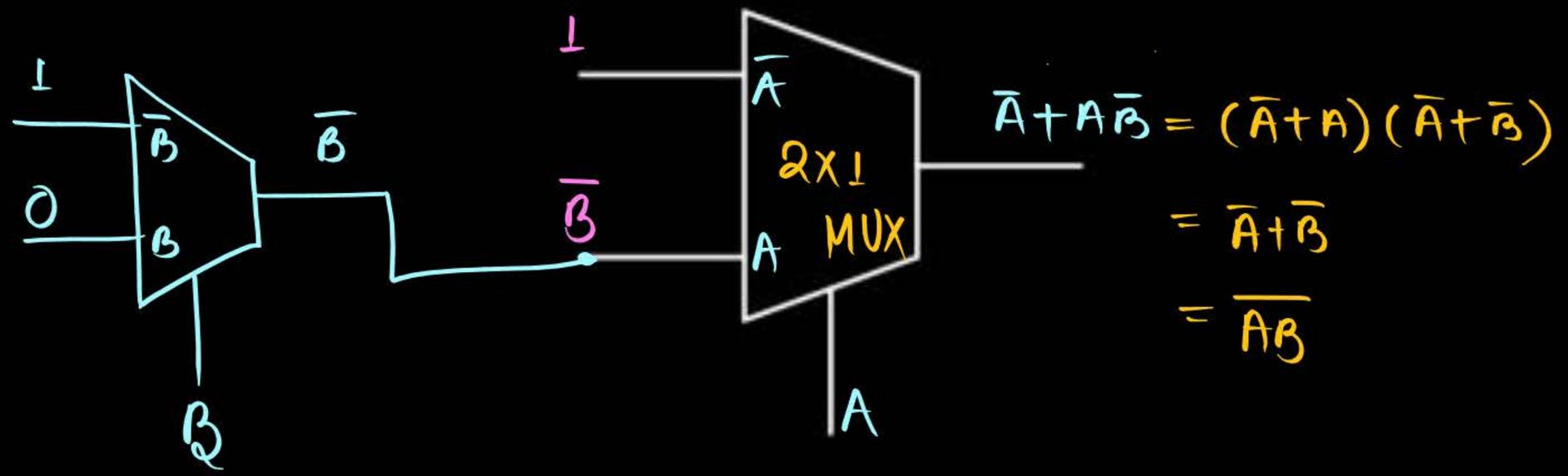




Type-2 : MUX as a Universal Logic

4. NAND GATE

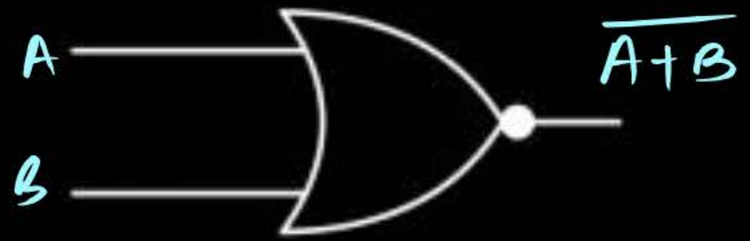


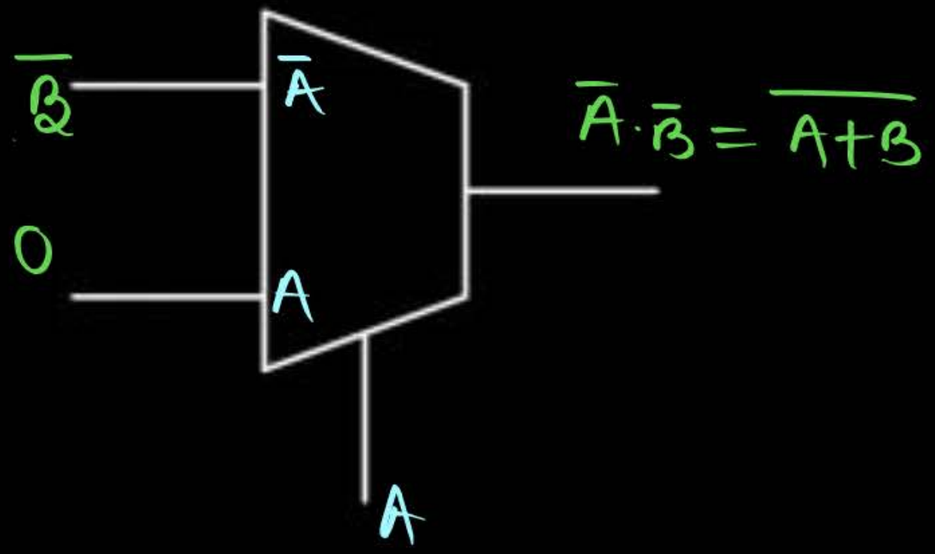




Type-2 : MUX as a Universal Logic

5. NOR GATE

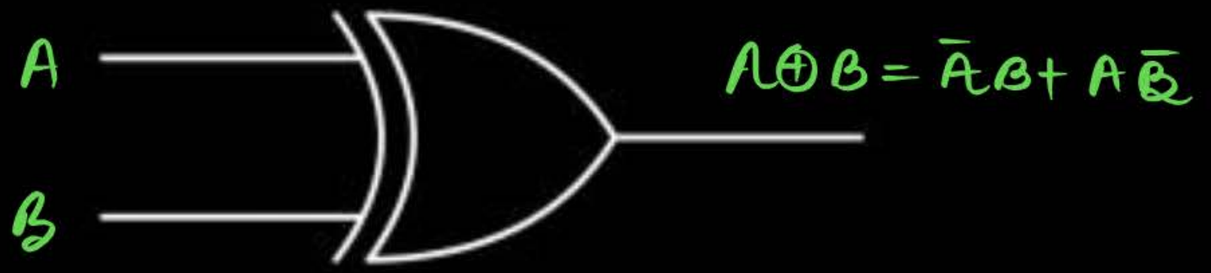


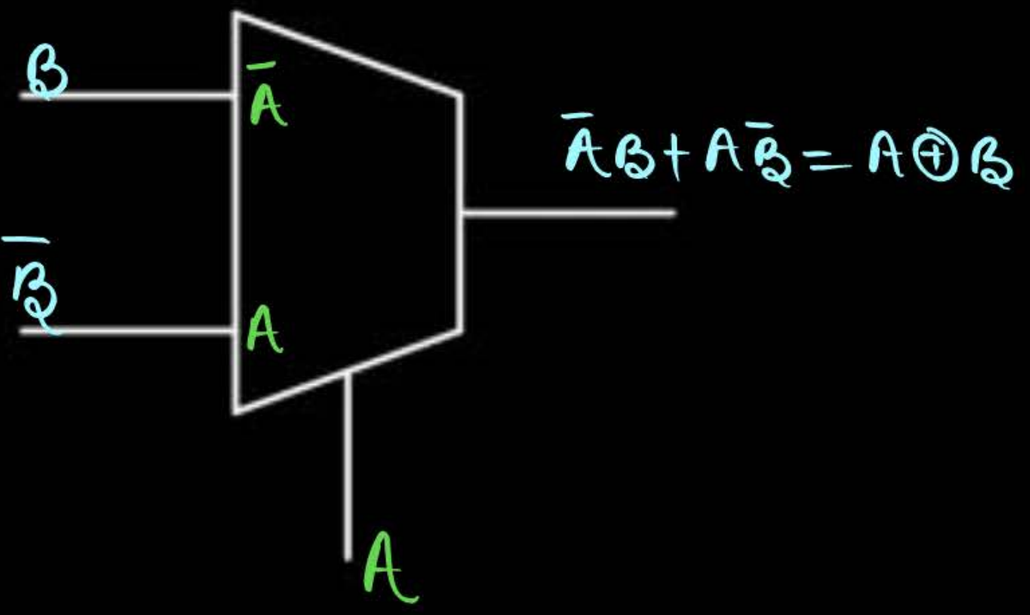




Type-2 : MUX as a Universal Logic

6. X-OR GATE

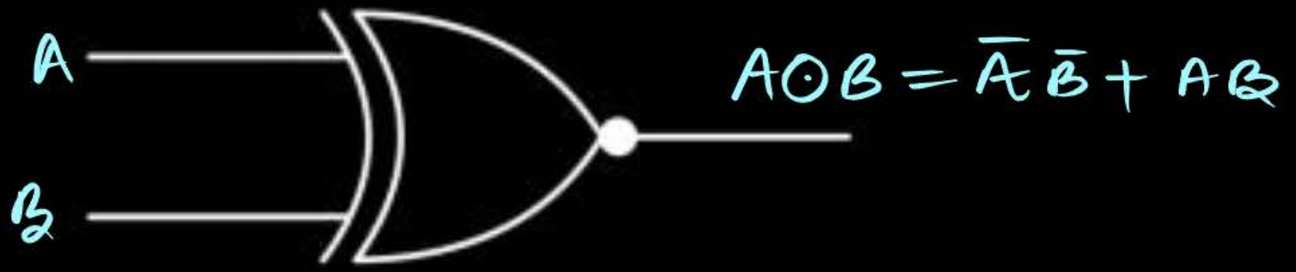


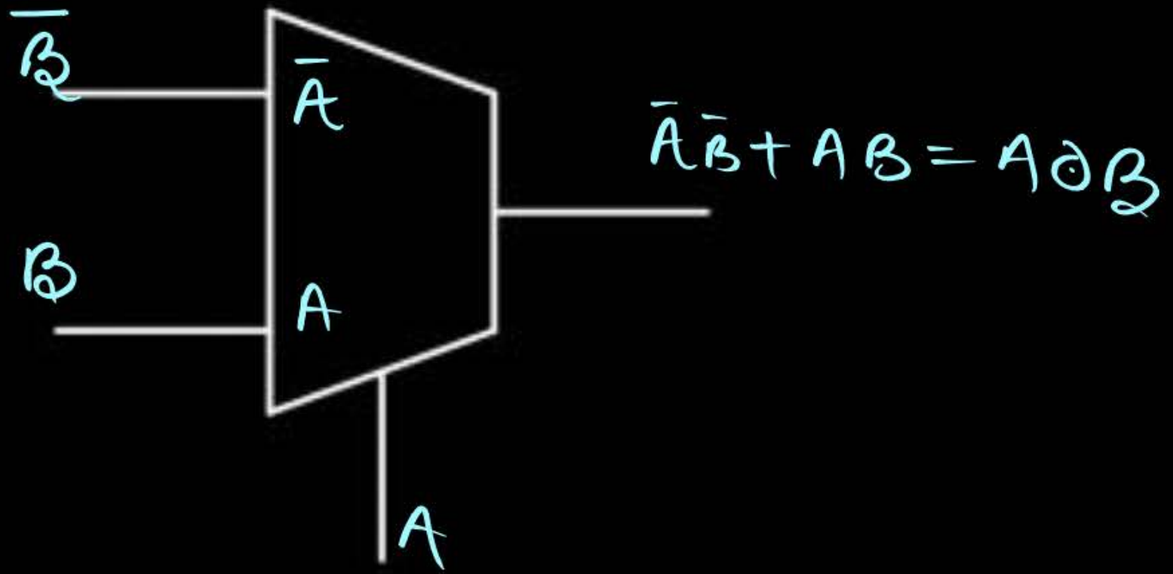




Type-2 : MUX as a Universal Logic

7. X-NOR GATE





NOTE

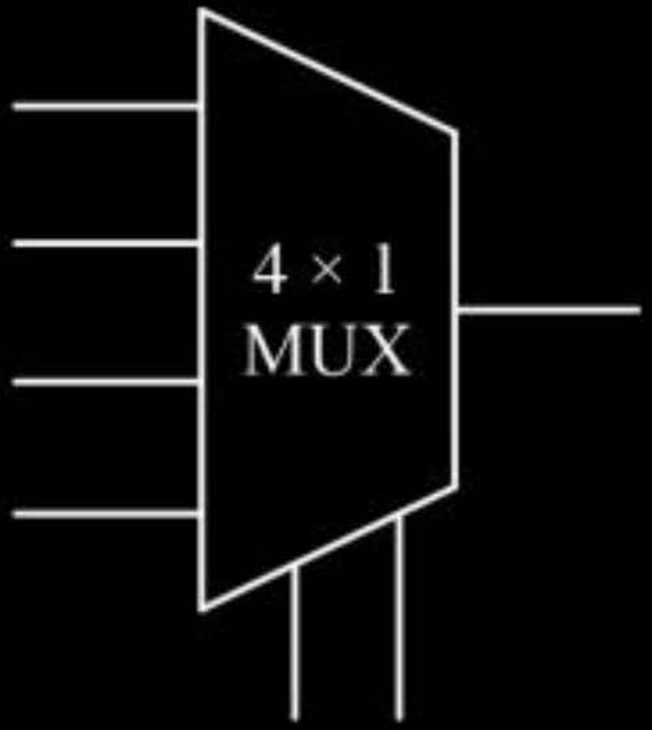
Σ^{10}

NOT, AND, OR $\Rightarrow$ One 2x1 MUX

NAND, NOR, X-OR, X-NOR $\Rightarrow$ Two 2x1 MUX

Type-3 : Minimization

#Q. Find the output f.



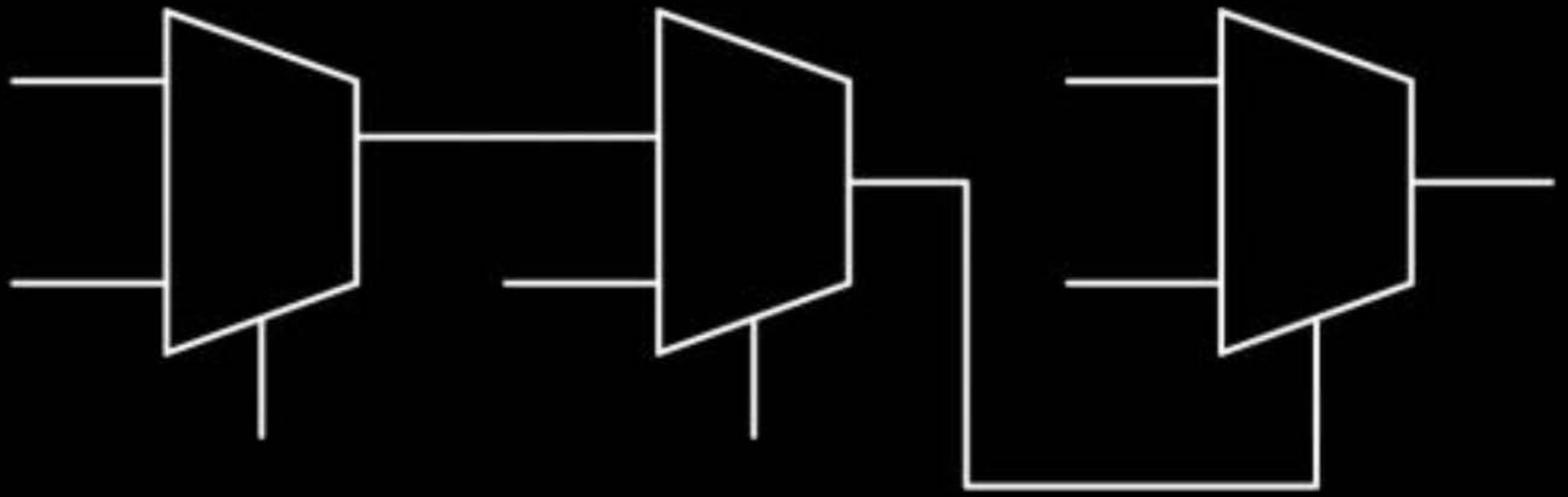


Type-3 : Minimization



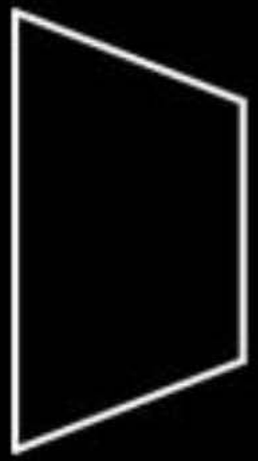


Type-4 : Cascading of Mux





Type-5 : Implementation of function





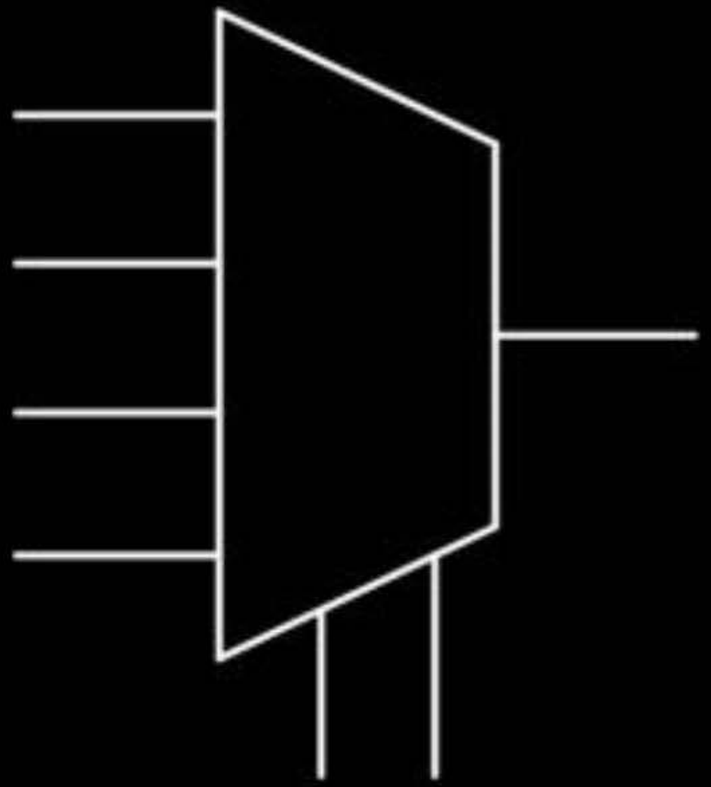


Topic : By using 4 × 1 MUX

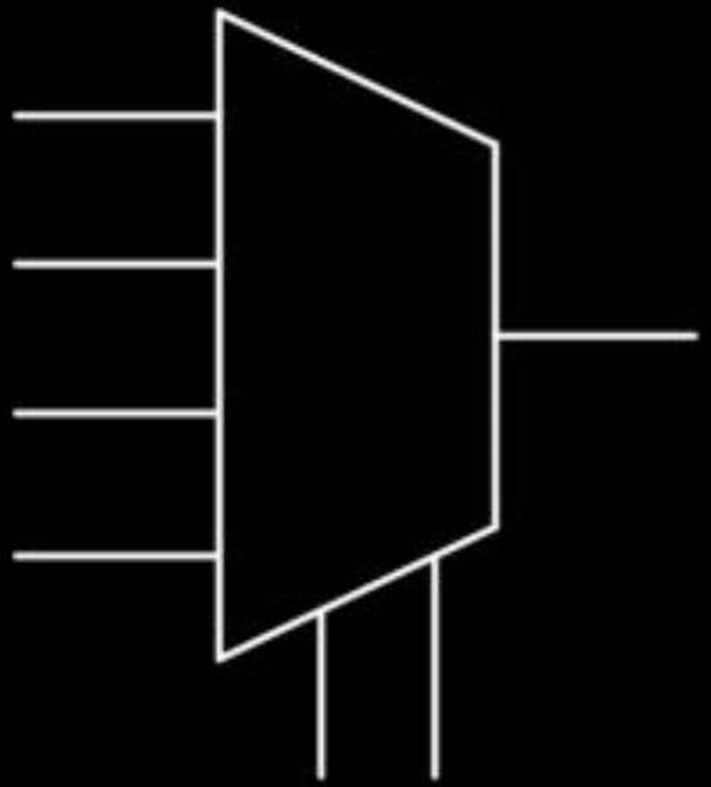
$$f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

1. AB as a select line
2. BC as a select line
3. AC as a select line

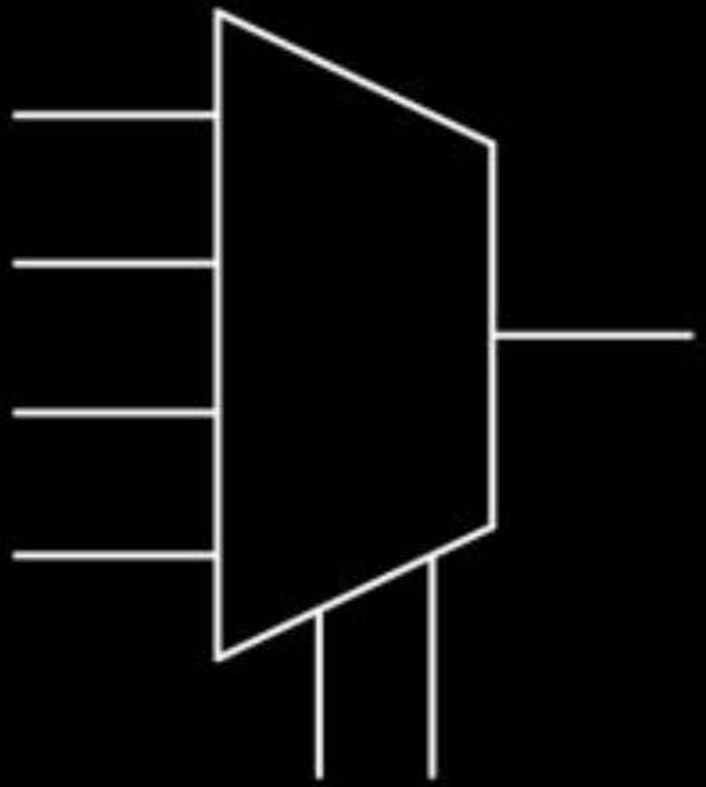
1. AB is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$



2. BC is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$

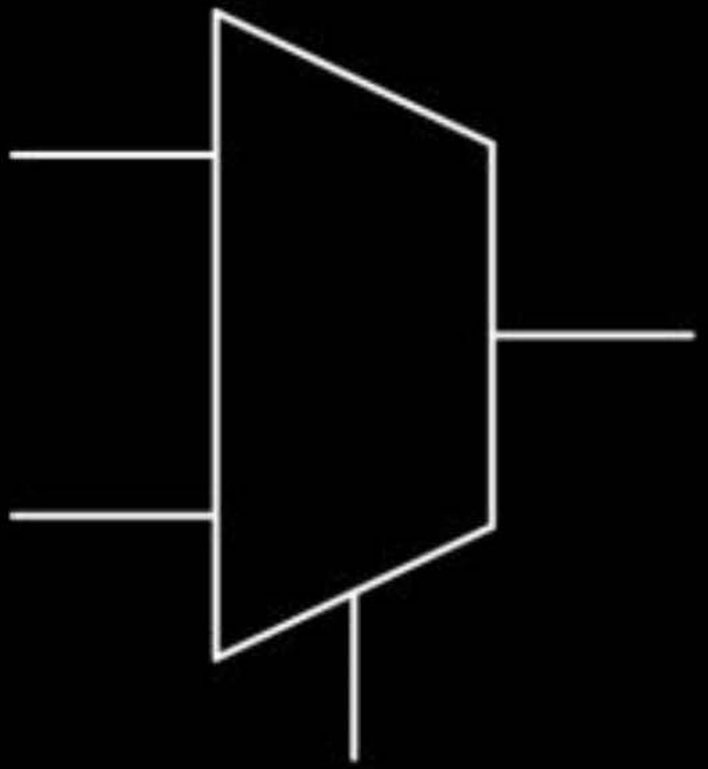


3. AC is select line $f(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$





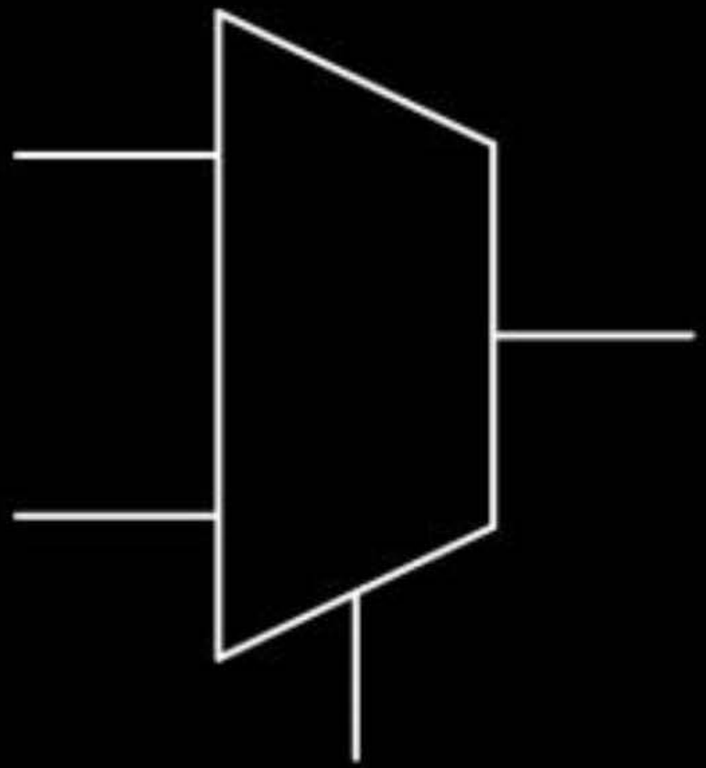
Topic : By using 2 × 1 MUX



#Q. $f(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 9, 12, 15)$

1. ABD as a select line
2. ACD as a select line

Type-6 : Delay



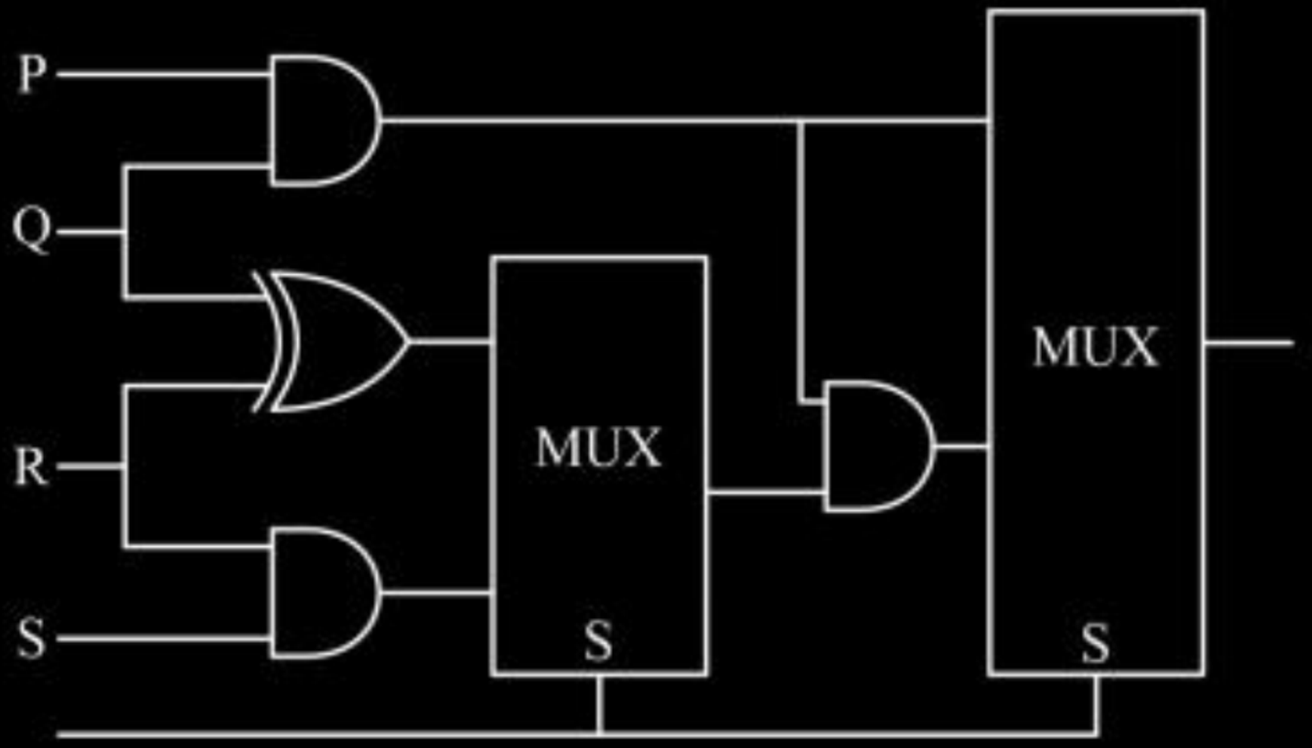
#Q. The propagation delay of the XOR gate, AND gate multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively.
If all the inputs P, Q, R, S and T are applied simultaneously and held constant the maximum propagation delay of the circuit is

A 3 ns

B 6 ns

C 5 ns

D 7 ns





2 mins Summary

MULTIPLEXER

QUESTIONS PRACTICE

DISCUSSION

Thank You GW Soldiers!

